

Quiz (Habanero)

- Q1.** A weather station records a -2.5 degree temperature drop. What is this drop as a fraction?
- (A) $-\frac{5}{2}$
(B) $-\frac{1}{2}$
(C) $\frac{1}{2}$
(D) $\frac{5}{2}$
(E) $-\frac{25}{10}$
- Q2.** What percent of the 24-hour day is the 4-hour period when pollen counts are highest?
- (A) 16.7%
(B) 20%
(C) 25%
(D) 33.3%
(E) 40%
- Q3.** In a certain ecosystem, 0.48 of the plants are mangroves. What fraction of the plants are not mangroves?
- (A) 0.52
(B) 0.48
(C) 0.51
(D) 0.49
(E) 0.53
- Q4.** Order the following rational numbers from least to greatest: -0.5 , $-\frac{1}{3}$, 0.2 . Which is the correct order?
- (A) $-0.5 < -\frac{1}{3} < 0.2$
(B) $-\frac{1}{3} < -0.5 < 0.2$
(C) $-0.5 < 0.2 < -\frac{1}{3}$
(D) $0.2 < -0.5 < -\frac{1}{3}$
(E) $-\frac{1}{3} < 0.2 < -0.5$
- Q5.** What is $4 \cdot 2.5$ in decimal form?
- (A) 9
(B) 10
(C) 11
(D) 12
(E) 10.0
- Q6.** A rain gauge measures 0.75 inches of rainfall. What is this as a percent of 3 inches?
- (A) 25%
(B) 30%
(C) 33.3%
(D) 40%
(E) 50%
- Q7.** Represent -0.25 on the number line. Which of the following is correct?
- (A) Between 0 and -1
(B) Between -1 and -2
(C) Between 0 and 1
(D) Between -0.5 and 0
(E) Between -0.5 and -1
- Q8.** A certain gas emits 2.1 tons of CO₂ daily. If it emits 0.5 tons more tomorrow, what is the increase as a fraction?
- (A) $\frac{1}{4}$
(B) $\frac{1}{2}$
(C) $\frac{1}{10}$
(D) $\frac{5}{10}$
(E) $\frac{21}{40}$

Open Ended Questions (FRQ)

- Q9.** The temperature of a city drops from 25 degrees to -5 degrees over 10 hours. At what rate does the temperature drop per hour as a decimal? Explain your steps and include your calculation.
- Q10.** Compare the fractions $\frac{1}{8}$, 0.15 , and $\frac{3}{20}$. Order them from least to greatest and explain your reasoning, showing your calculations. Be sure to represent your answer as a decimal for comparison.

Chef's Tips

- Tip 1: Ensure students bring a calculator and a pencil.
- Tip 2: Allocate exactly 45 minutes for this quiz.
- Tip 3: Review the conversion of fractions, decimals, and percents beforehand.